Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Total number of Rows: 3,900
- Total number of Columns: 18
- Key Features of the data:
- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Count of missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

The following defines the steps performed during the with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used df.head() to get an overview of the concerned data.

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Pay M
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	V
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	I
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	I

- df.info() to check structure of the data and the field types to verify the use of proper types in every column.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Customer ID             3900 non-null   int64  
 1   Age                     3900 non-null   int64  
 2   Gender                  3900 non-null   object 
 3   Item Purchased          3900 non-null   object 
 4   Category                3900 non-null   object 
 5   Purchase Amount (USD)   3900 non-null   int64  
 6   Location                3900 non-null   object 
 7   Size                    3900 non-null   object 
 8   Color                   3900 non-null   object 
 9   Season                  3900 non-null   object 
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object 
 12  Shipping Type           3900 non-null   object 
 13  Discount Applied        3900 non-null   object 
 14  Promo Code Used         3900 non-null   object 
 15  Previous Purchases      3900 non-null   int64  
 16  Payment Method          3900 non-null   object 
 17  Frequency of Purchases  3900 non-null   object 
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

- df.describe() for summary statistics.

```
df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

- Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- Column Standardization: Renamed columns to snake case for better readability and documentation.
- Feature Engineering:
  - Created age_group column by binning customer ages.
  - Created purchase_frequency_days column from purchase data.
- Data Consistency Check: Found that discount_applied and promo_code_used were redundant columns; hence dropped promo_code_used.
- Database Integration: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

Performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue and Order Summary by Gender – Compared total revenue generated by male vs. female
customers.

	gender text	total_revenue numeric	no_of_purchases bigint	avg_order_value numeric
1	Female	75191	1248	60.25
2	Male	157890	2652	59.54

Findings: Male customers purchase more than females while the avg order value is fairly the same.

2. High-Spending Discount Users – Identified high spending discount loving customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint	location text
1	2	64	Maine
2	3	73	Massachuse...
3	4	90	Rhode Island
4	7	85	Montana
5	9	97	West Virginia
6	12	68	Hawaii
7	13	72	Delaware
8	16	81	Rhode Island
9	20	90	Rhode Island
10	22	62	North Caroli...
11	24	88	Oklahoma
12	29	94	North Caroli

Total rows: 839 Query complete 00:00:00.136

Findings: Found customers who spend more when given discounts.

3. Top Rated Products – Found products with the highest average review ratings within each category and also checked if they are the top selling products.

Top Selling products:

	item_purchased text	overall_rating numeric	no_of_purchases bigint
1	Pants	3.72	171
2	Jewelry	3.76	171
3	Blouse	3.68	171
4	Shirt	3.62	169
5	Dress	3.75	166

Top Rated Products:

	item_purchased text	overall_rating numeric	no_of_purchases bigint
1	Gloves	3.86	140
2	Sandals	3.84	160
3	Boots	3.82	144
4	Hat	3.80	154
5	Skirt	3.78	158

Findings: The top rated products are not the top selling products however the top selling products also have fairly high ratings.

4. Comparing Shipping Options – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	avg_purchase_amount numeric
1	Standard	58.46
2	Express	60.48

Findings: The avg order value is slightly higher with the express shipping option.

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue between Subscribers and Non-Subscribers.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Findings: 27% of the customers have subscribed , the average order value is almost same among Subscribers and Non-Subscribers with the subscriber having slightly lesser avg order value. Subscribers have 26.88% contribution in the total revenue which is almost same as the percentage of subscribers among the customers.

6. Discount-Dependent Products – Identified products with the highest percentage of discounted purchases.

	item_purchased text	discount_frequency numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Findings: Found the products that heavily rely on the discounts for their sale.

7. Customer Segmentation – Classified customers into categories such as New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Findings: A large number of customers are loyal to the platform and we have a good number of returning customers. And the customer base is growing as the platform has some new customers as well.

8. Top Selling Products per Category – Listed 3 products with highest number of purchases within each category.

	item_rank bigint	category text	item_purchased text	no_of_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Findings: Found top 3 highest selling products in each category.

9. Repeat Buyers & Subscriptions Trend– Compared purchase history and subscription status to determine whether customers with more than 5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Findings: Our loyal customers are not subscribing as only 27.6% of our loyal customers have subscribed.

10. Revenue Segmentation by Age Group – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle_aged	59197
3	Adult	55978
4	Senior	55763

Findings: The 'Young Adult' section of our customer is the most active on our platform with the highest contribution in the revenue whereas the seniors have the smallest contribution among all the age groups. However the difference between the biggest and the smallest contribution is only $6380 where the largest contribution of 'Young adult section is only 11.44% more than the seniors .

5. Dashboard in Power BI

Built an interactive dashboard in Power BI to present insights visually. Used cards to represent the important numbers like the number of customers, average purchase amount, average review rating , total revenue.
Created charts to show the distribution of revenue, no. of sales across categories.
Added slicers to change the parameters for an interactive dashboard for better analysis.

6. Business Recommendations

● Boost Subscriptions – Promote exclusive benefits for subscribers to drive subscriptions.

● Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.

● Review Discount Policy – Balance sales boosts with margin control.

● Product Positioning – Highlight top-rated and best-selling products in campaigns to drive their sales.

● Targeted Marketing – Focus efforts on high-revenue age groups (Young Adults) also target females to drive to acquire new female customers .

● Focus on express-shipping methods to drive the average order value. .